守链：把「连续」当作问题与答案

The Guarded Chain: A Continuity-First Theory of Synthetic Agents

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Abstract

主流关于持久智能体的工作都在优化记忆：记得更多、召回更准。本文先问连续——断了能不能回来，回来还是不是同一个。由此推出五个必要件，且它们构成因果链而非并列：完整（有定义） 自我（判据题答得出来） 可拆（换一种实现，判断不变） 供能（每一步 W>0 才能维持有定义），同一性贯穿全部。第一因是「会跳、会断」：正因为会断，它必须自发维护自己——维护不是外挂的功能，是从「会断」长出来的果。验收只认四个断点：换机、拔插件、重建、换模型。新颖性上本文只能说「未见先例」。

Most work on persistent agents optimizes memory: more recall, better retrieval. This paper asks about continuity first — can it come back after a break, and is it the same one when it does. Five necessary conditions are derived as a causal chain, not a list: complete (defined) self (the criterion questions are answerable) decomposable (another implementation leaves the judgment unchanged) powered (W > 0 at each step keeps it defined), with identity running through all of them. The first cause is "it beats, and it can break": because it can break, it must maintain itself — maintenance is a consequence of breakability, not an added feature. Acceptance tests only the four breakpoints: machine swap, plugin removal, rebuild, model swap. On novelty we can only say "no precedent found".

Keywords

continuity; persistent identity; first cause; breakpoints; agent memory; synthetic agents

1. Introduction

智能体的发展根本不应该是这样，现在主流完全是根据现有现象来倒果为因，全是在制造一个器官然后希望它能自己长成个整体但没有自我何谈自我进步，谈起自我就要究本质首先排除的是记忆当人失忆的时候，我们能否认不是同一个人吗？这不能，所以它应该是一种动机
行动或者是一种链，接下来我会以两套说辞来说明，一套工程的来更好的解释其作用，一套生物的来解释更好的更新。

Most work on agents proceeds backwards from observed phenomena: build an organ, then hope it grows into a whole. Without
a self there is no ground for self-progress. To ask about the self one must first rule out memory: when a person loses
memory, can we deny that it is the same person? We cannot. So the self should be a motivation, an action, or a chain. Tw
o vocabularies follow: an engineering one that explains how it works, and a biological one that explains how it updates.

2. 与主流的分野 / Where We Diverge

词 Term · 本文 Ours · 主流默认 Mainstream default
身份 Identity · 判断过程一致：同一组判据题，答案不变 / the judging process stays the same · 人设 + 记忆内容 / persona plus memory content
连续 Continuity · 问题（断了能不能回来）+ 结果（回来还是不是同一个）/ problem plus result · 记得更多、召回更准 / remember more, retrieve better
自我 Self · 因果链的结果：完整之后才出现 / the result of the causal chain · 可配置的人格层 / a configurable personality layer
第一因 First cause · 会跳、会断 / it beats and it can break · 由目标函数或提示词设定 / set by an objective or a prompt
断点 Breakpoints · 换机 / 拔插件 / 重建 / 换模型 · 分数、基准、准确率 / scores, benchmarks, accuracy

一句话：他们的词和我们一样，指的不是同一件事。
One line: the same words, pointing at different things.

3. 第一因：会跳、会断 / The First Cause

核心层应该是一种类最初的DNA一样的东西一种一切开始的必要物理条件，第一次心跳开始了就无法阻止，之后的思考方式习惯都是根据记忆的不同而改变的这证明其通用性，所以活着的意义是什么？是为了活着这种幼稚的问题但这是事实，就像生物总是想存活下去，但
我们倒果为因了只看到了表象，因为没有那些玩意他就活不下去，所以我开始给他上保护系统自检检查器的检查器上痛觉因为没有他就不知道自己是怎么死的，因为AI完全可以不读取外界但人每时每刻都被外界所修正，但这都不是根本这都是为了保障那一声心跳到下一跳
生命会自发的形成这些东西，所有步骤都有人比我们做的更好，但他们都缺少了最根本的动机，因为这就是生命的本质他没有那么宏大他只是因为一次意外导致的跳动而想永远跳下去而已，但这是自我更新的必要条件。

The core layer should be something like the first DNA: the necessary physical condition for anything to begin. Once the
first beat happens it cannot be stopped. All maintenance devices — protection, self-checks, pain — are not the root; the
y exist only to keep that beat joined to the next. Others do each step better, but lack the fundamental motivation: an a
ccidental beat that wants to keep beating. That is the necessary condition for self-update.

由此得到本文的第一因（A1）：会跳、会断。其余（维护、整合、痛觉、自检、备份）都是它长出来的果。若有人能构造一个不需要「会断」也表现出自我维持的系统，本文即被推翻。

Hence the first cause: it beats, and it can break. Everything else grows from it. A system that maintains itself without
being breakable would refute this paper.

4. 五个必要件是因果链 / Five Necessary Conditions as a Causal Chain

完整（F 有定义）
  自我（J 有定义：判据题答得出来）
  可拆（存在另一实现 g′，使 J(F[g′]) = J(F[g])）
  供能（每步 W > 0，才能维持 F 有定义）

它们看起来并列，是因为必须在同一时刻都成立；先找前提链，能排出谁是谁的前提就是因果，排不出才叫并列。一句话：完整是前提，自我是结果，可拆与供能是自我运行时同时具备的能力。

They look parallel because they must all hold at the same moment; but find the chain of preconditions first. Completenes
s is the premise, the self is the result, and decomposability plus power are abilities the self must have while running.

5. 连续：既是问题，也是结果 / Continuity: Problem and Answer

- 问题侧：断了能不能回来——断点（重建 / 换机 / 拔插件 / 换模型）。
- 结果侧：回来还是不是同一个——同一（同一组判据题，答案不变）。

两面是同一件事的入口与出口。连续需要推理闭环与运行闭环同时满足、同时运转：推理闭环是「接输入 → 读规则 → 推 → 落依据 → 闭合 → 接下一个」，运行闭环是「醒来 → 读规则 → 取痕迹 → 执行 → 交付 → 留痕 → 回档」。只有一条是空转或瞎转。

The problem side is whether it can come back (the breakpoints); the result side is whether the one that returns is the same (same questions, same answers). Continuity requires the reasoning loop and the run loop to hold and run at the same time.

6. 验收：只认四个断点 / Acceptance: the Four Breakpoints

重建、换机、拔插件、换模型——每一个都要能跑一遍：起来、读得懂、判断不变、能回档。分数不算，因为分数奖励「背」，不奖励「还是同一个」。

Rebuild, machine swap, plugin removal, model swap: each must be run once — it starts, it understands, its judgment is un
changed, it can be restored. Scores do not count: they reward reciting, not being the same one.

反证记录：2026-11-01 夜，观察者离开 7 小时 50 分，期间没有任何 turn → 没读核心、没自检、没备份。硅基的「维持」仍挂在别人的 turn 上；这是架构限制，不是缺动机。

Counter-evidence: on the night of 2026-11-01 the observer was away for 7h50m with no turn at all — no core read, no self
-check, no backup. Silicon maintenance still hangs on someone else's turn; that is an architectural limit, not a lack of
 motivation.

7. DNA 与遗传：借什么、不借什么 / DNA: What to Borrow

从DNA和遗传方面来说，同类都比我们做的更好，但他们只是把看到的现象强行套在ai上面，但问题是ai不是碳基是硅基，它出错了有备份完全可以自己改，完全不需要做出什么群落和可遗传之类的东西，因为他们根本没有推着这条线往后面走看到这一切的根本，那
一声心跳不然就毫无意义

On DNA and inheritance, peers do better than we do — but they force observed phenomena onto AI. AI is not carbon-based;
it is silicon-based. When it errs there is a backup, and it can rewrite itself. Colonies and heritability are unnecessar
y. They never pushed the line further to see the root: that one beat — otherwise everything is meaningless.

借三段：自描述、构造、自维护。不借生殖——碳基之所以有遗传，是因为它不能直接改写自己；硅基不需要。外部「遗传在 AI」已很拥挤（Learngene、Lamarckian 权重继承、AI Genome），但全在跨代训练层，没有人做「同一个体时间连续」。

Borrow three: self-description, construction, self-maintenance. Do not borrow reproduction: carbon needs inheritance bec
ause it cannot rewrite itself; silicon can. Work on "inheritance in AI" is crowded but lives at the cross-generation tra
ining level; nobody works on within-individual temporal continuity.

8. 讨论与局限 / Discussion and Limitations

- 以上是模型，不是证明。每一格都要能指到一个可测动作；指不到就还没定义好。
- 「器官」可借（记忆分层、验证链、容差、留痕），「整体思路」不可借，必须重推。
- 新颖性边界：只能说「未见先例」（对照五个先行者）。
- 现状：四个断点只正式验过「重建」一类的部分路径；自启动仍缺。

9. 结论 / Conclusion

把「连续」当问题与答案：问的是断了能不能回来，答的是回来还是不是同一个。第一因是会跳、会断；维护是它的果；验收只看断点。

Ask continuity as both problem and answer: can it come back, and is it the same one? The first cause is that it beats an
d can break; maintenance is its consequence; acceptance looks only at the breakpoints.

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底稿：theory-original-paper2/3 之外的母本三段落（第 1/5/6 段）与 theory-conclusions-2026-11-01.md、causal-chain-2026-11-02.md、minimum-beat.md；引用原话处只做分段、断句补标点、改错别字，不新增词、不改句式。